

Cognitive Security as a Technical Problem

Assistance Timing, Removed-Tool Evaluation, and Two Catalog Entries for AI-Mediated Learning

Hao-Tian Lu¹ · Kuan-Wei Lu¹

¹Junyi Academy Foundation, Taipei



CogSec '26 · Alexandria, VA · 27 September 2026

The threat

AI-mediated learning is a cognitive-security threat with the same protected object as adversarial influence but a different structure: non-adversarial, cumulative, *constitutive*, not distortionary.

Pre-attempt assistance doesn't reduce the learning signal — it removes it.

The mechanism

$$\delta = \mathcal{L}(\hat{y}_L, y)$$

Post-attempt sharpens δ . **Pre-attempt** supplies y before $\hat{y}_L$ exists, so δ is *unavailable*, not small.

No free parameter for how much help — the variable is *when*.

The axis

Two links break at once: competence is no longer built by doing, and performance no longer reveals it.

Over- and under-reliance are symmetric failures of one calibration.

Four independent lines of evidence

Learning harm. Math RCT, ~1,000 students: exam scores fall once the tool is withdrawn.

Neural signature. EEG during essay writing: connectivity drops with support, lowest ownership.

Critical engagement. Trusting the AI more meant scrutinizing it less; trusting oneself more meant scrutinizing it more.

Perception gap. Developers measurably *slowed* by AI tooling, yet felt sped up.

Two draft entries for the CVMC

E1

Pre-attempt answer generation

Affordance	complete solution on demand
Vulnerability	no expectation is formed
Failure pattern	practice rises, retention falls
Mitigation	withhold, then critique
Metric	decay on withdrawal
Burden check	time, language, processing
Governance hook	procurement disclosure

E2

Miscalibrated friction

gate until first attempt
support refused or abandoned
compliance recorded, support lost
calibrate + accommodation path
declined-assistance rate
disaggregation; non-stigmatising
audit rights; disable/recalibrate

Pre-attempt answer generation leaves no expectation to violate; withhold until the learner commits, and measure with the tool withdrawn.